EXERCISE 4 SOLUTIONS

1a) $H_0: \mu = 15.9$

$H_a: \mu \neq 15.9$

b) $\alpha = 0.05$

c) USE $t = \dfrac{\bar{x} - \mu}{s/\sqrt{n}}$ SINCE SAMPLE STANDARD DEVIATION IS KNOWN

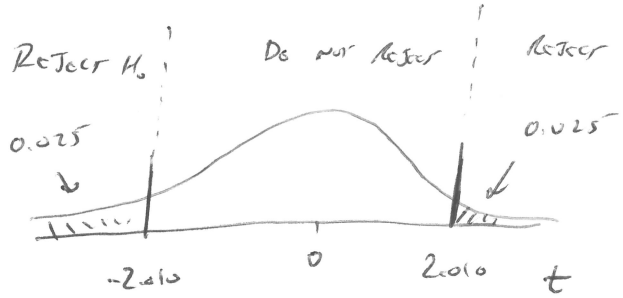
d) S1: H_a "$\neq$" $\rightarrow$ 2-TAILED TEST.

S2 $\alpha = 0.05 \rightarrow \alpha/2 = 0.025$

S3 USE t

$n = 50$, $df = 50 - 1 = 49$

$t_{0.025,49} = 2.010$



DECISION RULE: REJECT H_0 IF $t_{CALC} < -2.01$ OR $t_{CALC} > 2.01$

e) $\bar{x} = 17.8$

$s = 6$

$n = 50$

$\mu = 15.9$ (STEP 1)

$$t_{CALC} = \frac{17.8 - 15.9}{6/\sqrt{50}}$$

$$t_{CALC} = 2.2392$$

SINCE $t_{CALC} = 2.2392 > 2.01$, WE REJECT H_0.

THERE IS SUFFICIENT EVIDENCE TO SUGGEST THAT MEAN LENGTH OF IMPRISONMENT IN TORONTO DIFFERS FROM THE NATIONAL AVERAGE

2a) $\mu = 15.9$ IS OUTSIDE OF THE CONFIDENCE INTERVAL RANGE

(16.09, 19.51)

b) $\mu = 15.9$ IS OUTSIDE THE INTERVAL AND WE REJECTED THE NULL HYPOTHESIS IN Q1, THIS IS CONSISTENT WHEN THE α IS THE SAME BETWEEN A CONFIDENCE INTERVAL (eg. 95% => $\alpha = 0.05$) AND LEVEL OF SIGNIFICANCE FOR A TEST, <u>AS LONG AS</u> THE HYPOTHESIS TEST IS 2-TAILED.